

Zekun Wang

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Research Focus

I study continual learning for deployment-time adaptation and generalization in foundation models and AI agents. I investigate how these systems can learn from ongoing experience, adapt to unfamiliar tasks and environments, and improve over time while preserving prior knowledge and capabilities.

Education

Georgia Institute of Technology 2024 – 2029

Ph.D. in Computer Science

Advisor: Christopher J. MacLellan

University of Michigan 2022 – 2024

M.S. in Computer Science

Pennsylvania State University 2018 – 2022

B.S. in Computer Science

Double major: Mathematics. Minor: Economics.

Industry Experience

Amazon 08/2026 – present

Research Intern · WA, USA

- **One-shot adaptation to novel tasks.** Developing methods that adapt an agent to an unfamiliar task at deployment through a single update to its harness.
- **Self-improving agent harnesses.** Contributing to a harness self-improvement system that outperforms baseline harness-evolution methods on SWE-bench and Terminal-Bench.

Selected Publications

For the full publication list, see [Google Scholar](#).

* Equal contribution.

Self-Consolidating Language Models: Continual Knowledge

Preprint

Incorporation from Context

Zekun Wang^{*}, Anant Gupta^{*}, Zihan Dong, Christopher J. MacLellan

- Developed a meta-RL post-training framework in which an LLM learns which layers to update as new context arrives, balancing knowledge acquisition against forgetting.
- Improved SQuAD acquisition and retention over sequential SEAL, and LongBench v2 accuracy over sequential fine-tuning; learned update policies transferred to longer context streams.

Avoid Catastrophic Forgetting with Rank-1 Fisher from Diffusion

ICLR 2026

Models

Zekun Wang^{*}, Anant Gupta^{*}, Zihan Dong, Christopher J. MacLellan

- Derived and empirically validated a low-SNR rank-1 Fisher approximation near convergence, enabling structured EWC with generative replay.
- Reduced mean FID-based forgetting by **67% on ImageNet-1k** (32×32, 20 tasks) versus generative distillation alone; nearly eliminated forgetting on MNIST and FashionMNIST.

Test-Time Compositional Generalization in Diffusion Models via Concept Discovery

Preprint

Zekun Wang*, Anant Gupta, Tianyi Zhu, Christopher J. MacLellan

- Developed test-time concept discovery and composition from a pretrained diffusion model's scores, enabling novel combinations **without retraining the base model**.
- Outperformed query-only and nearest-class baselines on ColorMNIST and CelebA composition using a product-of-experts sampler and a distilled variant.

Trust Region Continual Learning as an Implicit Meta-Learner

Preprint

Zekun Wang*, Anant Gupta*, Christopher J. MacLellan

- Proved that replay with a Fisher-metric trust region yields a one-step MAML-like update under local approximations, connecting continual learning to meta-learning.
- Improved retention and re-convergence over EWC, replay, FTML, and VR-MCL on ImageNet-500 diffusion generation and Continual-World-10 diffusion policies.

Deep Taxonomic Networks for Unsupervised Hierarchical Prototype Discovery

NeurIPS 2025

Zekun Wang, Ethan Haarer, Tianyi Zhu, Zhiyi Dai, Christopher J. MacLellan

- Developed a hierarchical generative model that discovers prototypes without a known class count and achieved state-of-the-art hierarchical prototype discovery in the paper's evaluation.
- Proved that its variational objective also optimizes mutual information between latent representations and the hierarchy.

Research Experiences

Georgia Institute of Technology

Graduate Research Assistant — Teachable AI Lab

Aug. 2024 – present

University of Michigan

Graduate Research Assistant — SLED Lab (PI: Joyce Chai)

2022 – 2024

Pennsylvania State University

Undergraduate Research Assistant

2021 – 2022

Teaching Experiences

Graduate Student Instructor — EECS 595, Natural Language Processing

University of Michigan

Sept. 2023 – May 2024

Teaching Assistant — CMPSC 464, Introduction to the Theory of Computation

Pennsylvania State University

Jan. 2022 – May 2022

Teaching Assistant — MATH 230, Calculus and Vector Analysis

Pennsylvania State University

Aug. 2021 – Dec. 2021

Math Peer Tutor — Calculus

Penn State Learning

Jan. 2020 – Jan. 2022

Invited Talks and Guest Lectures

Intro to AI (CS 3600), Georgia Tech — Guest lecture

Aug. 2025

Deep Learning: Classics and Trends (DLCT) — Invited talk

Babysit a Language Model From Scratch: Interactive Language Learning by Trials and Demonstrations

July 2024

Advanced Artificial Intelligence (EECS 692), U. of Michigan — Guest lecture

Jan. 2023

Academic Leadership and Service

Workshop Organizer

Towards Test-Time Continual Learning Agents (TTCL), NeurIPS 2026

Dec. 2026 (*upcoming*)

Student Volunteer: ACS.

Conference Reviewing: NeurIPS, ICML, ICLR, AAAI, ACL, NAACL, ACS.

Research Mentoring

Anant Gupta: M.S., Georgia Tech. 2025 – present.

Karthik Singaravadivelan: B.S., Georgia Tech. 2025 – present.

Aryan Roy: M.S., Georgia Tech. 2025 – present.

Zihan Dong: M.S., Georgia Tech. / Lenovo. 2024 – present.

Ethan Haarer: M.S., Georgia Tech. 2024 – 2025.

Skills

Programming	Python, C/C++, CUDA, R, MATLAB, Java, C#, $\text{\LaTeX}$, SQL
ML Frameworks	PyTorch, Hugging Face (Transformers, Diffusers, PEFT/LoRA, TRL), Accelerate, DeepSpeed, vLLM, JAX (working), Weights & Biases
Infrastructure	SLURM, Docker, multi-node A100/H100 clusters, Linux/Bash, Git
Mathematics	Information / probability theory, Bayesian inference, calculus, linear algebra, variational methods, score matching, optimization
Research areas	Continual / lifelong learning, LLM post-training (RLHF/PPO/DPO/meta-RL), diffusion & flow models, meta-learning, compositional generalization, concept formation
Languages	English (fluent), Cantonese (fluent), Mandarin Chinese (native)

Full Publication List

Diffusion Models and Concept Formation Zekun Wang, Karthik Singaravadivelan, Christopher J. MacLellan <i>arXiv preprint</i>	2026
Decompose Sparsely Where You Should, Absorb Densely Where You Should Not Ruixuan Deng, Zehao Jin, Zekun Wang, Zihan Dong <i>ICML 2026 Mechanistic Interpretability Workshop (Poster)</i>	2026
CobwebTM: Probabilistic Concept Formation for Lifelong and Hierarchical Topic Modeling Karthik Singaravadivelan [*] , Anant Gupta [*] , Zekun Wang, Christopher J. MacLellan <i>Findings of ACL 2026</i>	2026
Grounded Concreteness: Human-Like Concreteness Sensitivity in Vision-Language Models Aryan Roy, Zekun Wang, Christopher J. MacLellan <i>Findings of ACL 2026</i>	2026
A Rational Account of Categorization Based on Information Theory Christopher J. MacLellan, Karthik Singaravadivelan, Xin Lian, Zekun Wang, Pat Langley <i>CogSci 2026</i>	2026
Test-Time Compositional Generalization in Diffusion Models via Concept Discovery Zekun Wang, Anant Gupta, Tianyi Zhu, Christopher J. MacLellan <i>arXiv preprint</i>	2026

Self-Consolidating Language Models: Continual Knowledge Incorporation from Context Zekun Wang [*] , Anant Gupta [*] , Zihan Dong, Christopher J. MacLellan <i>arXiv preprint</i>	2026
Avoid Catastrophic Forgetting with Rank-1 Fisher from Diffusion Models Zekun Wang [*] , Anant Gupta [*] , Zihan Dong, Christopher J. MacLellan <i>ICLR 2026</i>	2026
Computer Vision Modeling of the Development of Geometric and Numerical Concepts in Humans Zekun Wang , Sashank Varma <i>AAAI 2026</i>	2026
Trust Region Continual Learning as an Implicit Meta-Learner Zekun Wang [*] , Anant Gupta [*] , Christopher J. MacLellan <i>arXiv preprint</i>	2026
Deep Taxonomic Networks for Unsupervised Hierarchical Prototype Discovery Zekun Wang , Ethan Haarer, Tianyi Zhu, Zhiyi Dai, Christopher J. MacLellan <i>NeurIPS 2025</i>	2025
Hierarchical Semantic Retrieval with Cobweb Anant Gupta [*] , Karthik Singaravadelan [*] , Zekun Wang <i>Advances in Cognitive Systems 12 (2025)</i>	2025
Efficient and Scalable Masked Word Prediction Using Concept Formation Xin Lian [*] , Zekun Wang [*] , Christopher J. MacLellan <i>Cognitive Systems Research 92, 101371</i>	2025
Taxonomic Networks: A Representation for Neuro-Symbolic Pairing Zekun Wang , Ethan L. Haarer, Nicki Barari, Christopher J. MacLellan <i>NeuS 2025</i>	2025
Computer Vision Models Show Human-Like Sensitivity to Geometric and Topological Concepts Zekun Wang , Sashank Varma <i>CogSci 2025</i>	2025
Babysit A Language Model From Scratch: Interactive Language Learning by Trials and Demonstrations Ziqiao Ma [*] , Zekun Wang [*] , Joyce Chai <i>NAACL 2025</i>	2025
Has It All Been Solved? Open NLP Research Questions Not Solved by Large Language Models Oana Ignat [*] , Zhijing Jin [*] , Artem Abzaliev, Laura Biester, Santiago Castro, Naihao Deng, Xinyi Gao, Aylin Gunal, Jacky He, Ashkan Kazemi, Muhammad Khalifa, Namho Koh, Andrew Lee, Siyang Liu, Do June Min, Shinka Mori, Joan Nwatu, Veronica Perez-Rosas, Siqi Shen, Zekun Wang , Winston Wu, Rada Mihalcea <i>LREC-COLING 2024</i>	2024
Foundations for AI-Assisted Formative Assessment Feedback for Short-Answer Tasks in Large-Enrollment Classes Susan E. Lloyd, Matthew D. Beckman, Dennis K. Pearl, Rebecca J. Passonneau, Zhaohui Li, Zekun Wang <i>ICOTS 2022</i>	2022